Spatialised sound: the listener's perspective[1]

Peter Mcilwain

School of Music – Conservatorium
Monash University

Peter.Mcilwain@arts.monash.edu.au

This paper compares the listener's perceived differences, in terms of space, between sound created by acoustic instruments and loudspeakers. The argument is made that, unlike acoustic instruments, sound from loudspeakers is not perceived as existing in the same personal space as that of the listener. Given that the large majority of music is now heard via loudspeakers, the paper discusses the possible implications in cultural terms. Following on from this is a discussion of some of the potential solutions to the spatial limitations of loudspeakers that have been developed. These include the creation of surround sound systems and virtual sound illusions via signal processing. Conclusions are made as to possible developments that could occur in the area of spatialised sound via loudspeaker playback.

Keywords: Sound Spatialisation, Loudspeakers, Ambisonic, Acousmatic Music, Sound Localisation

Introduction

Electronic music has, in the large majority of cases, only one actual instrument – the loudspeaker. This instrument is remarkable in its ability to create an enormous range of sounds. In fact so much so, that loudspeakers are often wrongly assumed to be able to create, or reproduce, any sound at all. For musicians and composers, loudspeakers can become conceptually invisible in the creative chain. This is despite the fact that loudspeakers do have limitations, particularly in relation to spatial imaging and, that these limitations strongly influence the type of music created for them.

- If we think about the spatial element in sound from the listener's perspective, then there are four main factors to consider:
- the sound itself,
- the environment in which the sound is transmitted,
- the spatial cues encoded in the sound[2],
- the correlation between spatial cues and the listener's ability to identify and position the sound source.

These factors affect the way a listener will perceive the spatial element of the sound and how they might relate this to a musical context. In considering this, it is first necessary to consider the sound sources that are under discussion here.

All physical objects, such as musical instruments and loudspeakers, must operate in an air-filled space (referred to here as *acoustic space*). As electronic music is almost completely reliant on the loudspeaker, it is useful to understand the relationship of the four elements of spatiality mentioned above, to the behaviour of loudspeakers in an acoustic space.

Furthermore, an understanding of the differences between musical instruments and loudspeakers, in these terms, seems desirable as many of the conventions and culturally-based interpretations of music stem from our tradition of music played by instruments.

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[1] This paper is an adapted version of sections from the masters thesis "An Approach to Spatio-temporal Patterning in Computer Music" Peter McIlwain, 2000.

[2] For a detailed discussion of sound localisation, see Kendall (1995).

One of the major differences between musical instruments and loudspeakers is that loudspeakers are able to create spatial cues from a different environment to that of the acoustic space of the listener. In the case of headphones, these spatial cues are, in the main, not modified by the listener's acoustic space. Here the listener can become immersed in a different acoustic space to the one in which they are situated. This is not a real space, in that it is not the result of the physical environment in which the listener is in. For this reason these spaces are described here as *virtual spaces*. In addition to the discussion outlined above, this paper will discuss the implications for music presented in virtual space compared to music that occurs in the physical environment.

A Comparison Between Loudspeakers and Musical Instruments

As discussed above, both loudspeakers and musical instruments operate in acoustic space. Their interaction with acoustic space, and the way in which the listener decodes the spatial information from these two types of sound sources, can be different. The spatial characteristics of sound coming from an acoustic instrument tends to be perceived by the listener in such a way that it is directly linked with the acoustic space in which they and the instrument reside. In other words, the sound is perceived as being in the listener's personal space. On the other hand, sound from loudspeakers tends to be heard as coming from somewhere else, or at least as not being part of the listener's acoustic environment. This perceptual difference may be caused by three factors:

- the difference between the ways in which the two types of instruments radiate sound,
- the fact that the playback of recorded sound through loudspeakers carries spatial information from another acoustic space[3], whereas acoustic instruments do not, and,
- the difficulty that listeners may have in creating imagined sound sources, and therefore sound loci, for electronically-generated sounds.

Acoustic instruments generally project sound in all directions. Their radiation patterns are typically complex as the pattern changes depending on the frequency that is emitted by the instrument. It is this complexity, together with the resultant reverberation caused by the interaction of the environment and the projected sound, that gives listeners clues as to the identity of the sound and its position in space (Roads, 1996).

Loudspeakers, on the other hand, radiate sound in a forward projecting *radiation pattern*. In the case of the loudspeaker, the listener is mainly hearing (in optimal listening conditions) the direct sound coming from the speaker. In contrast, when listening to an acoustic instrument, the sound of the instrument becomes part of the acoustic space in which it is created. This is due to the fact that the listener is hearing direct sound, as well as the complex interaction of that sound with the acoustic environment. On the other hand, complex interactions between the environment and sound coming from loudspeakers are more likely to distort, or interfere with, the intended sound. Furthermore, radiated sound from loudspeakers carries fewer clues as to the identity and localisation of the sound. For this reason locatisation cues are often included in the audio signal (such as artificial reverberation). However this solution gives rise to the second problem where the sound of a virtual space is superimposed upon the acoustic space.

A third factor comes into play when projecting sound over loudspeakers which is related to ability of a listener to create what Wishart (1996, pp 129 –161) calls *a sound landscape*. In his extensive discussion on the spatial properties of sound he states that the mental apparatus concerned with hearing is predisposed to allocating sounds to sound sources. This happens naturally in the performance of music played by acoustic instruments where the sounds are sourced to their physical origin. Listeners are able to do this by correlating the direct and indirect sound of the instrument with the visual stimulus of the performers playing.

[3] An exception to this is recordings that are made with microphones in the same space and position as the loudspeakers that playback the recording.

If however a recorded performance of acoustic instruments is played over loudspeakers the physical location of the sound is the loudspeakers themselves. In this instance, according to Wishart, the listener imagines the spatial and acoustic environment, or sound landscape, of the recording. The listener is able to do this because the sounds, and the mechanisms that generated them, are known. This is important because the listener is able to relate the behaviour of the sound of the instrument with that of the environment (such as reverberation or contextualising sound cues that may help to determine the nature of the acoustic space).
Sounds from synthesisers, on the other hand, are harder (or even impossible) to source, even in an imaginary sense. This is because they do not have a physical source. A listener therefore, may not be able to imagine a landscape for synthesised sounds as they have no physical origin. This can also apply to recorded sounds that are difficult to source or ambiguous as to their identity. For this reason, sounds of this type projected over loudspeakers may sound detached from the listener's acoustic space and even their imagined spaces.

Virtual and Acoustic Spaces with Stereo Loudspeakers

The difference in the spatial qualities of the two types of instruments is particularly marked where stereo sound recordings are played over stereo loudspeaker systems. These recordings carry the spatial characteristics of the environment in which they where recorded. As discussed above this sound exists in a virtual space. When the recording is played, it must be transmitted through the acoustic environment of the listener. This can result in a distortion, or even collapse of, the virtual space, as the two different spaces are overlayed. In order to avoid this, loudspeakers are often utilised in such a way as to negate the acoustic properties of the space in which they are in. One way that this is done is use loudspeakers in acoustic spaces that have high acoustic absorption characteristics and thus minimal impact as an acoustic space.

Listening to a recording produced by loudspeakers is the perceptual equivalent of looking at photographs. The sound is a kind of imported reality or the illusion of reality (MacDonald, 1995). If a sound is imported into a listener's 'reality' and is therefore not part of their perceived acoustic space, then, by inference, it must come from somewhere else.

This observation, in relation to loudspeakers, has been noted by Moore (1989) in his paper "Spatialization of Sounds over Loudspeakers". Here he describes the experience of listening to loudspeakers as perceptually equivalent to a situation in which a listener, in a room, is hearing a sound through a hole in the wall. An everyday example of this is when a listener in a room opens a window. The sound that is heard coming through the window is from another acoustic environment and is heard as such.

It is important to note here that Moore's observations are not just supposition. He is able to show that his model of perceived spatiality in relation to loudspeakers works by giving an algorithm that can create spatial effects. The algorithm works by calculating differences in onset times and relative amplitudes for paths travelled by sound through two "holes in a wall" (see Figure 1).

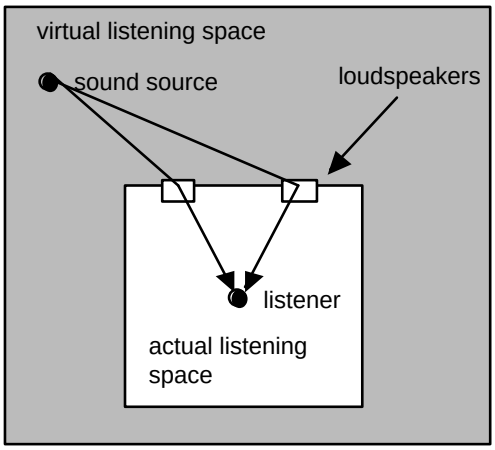


Figure 1. Moore's algorithm for spatialiation that makes a distinction between the ‘virtual’ and ‘actual’ listening spaces.

The Spatial Illusion and Listening Culture

The development of audio recording and reproduction technology introduced a radical change in the perception of music. It was now possible to listen to music that was not created in the same time and space as that of the listener. The modern experience tends to take this fact for granted even though (or possibly because) it is now the case that most music is heard through loudspeakers. The tendency to relate, at an unconscious level, loudspeakers as the producers (or reproducers) of “sound which comes from somewhere else”, has become part of the culture of listening to music produced by electronic means.

This culture now embraces not only spatial illusions which are intended to reproduce real acoustic spaces, but illusions of space which would not be possible in the real world. An example of this is exemplified in one approach to the recording of orchestras. Here a balance between the various instruments is created which is not necessarily intended to represent what is heard in the concert hall. Instead the balance is created such that all of the instruments can be heard clearly. Instrumental parts that might be masked when heard in the concert hall, due to the limitations of the power of the instruments concerned, can be compensated for in the recording process. This type of balance could only be achieved in a real world equivalent if the listener were able to be in several places at the same time.

Popular music, which is often created in the recording studio, commonly uses unreal acoustic spatial illusions. For example, a vocal line can be treated in such a way that it appears that the singer is enormous in size (as a spatial illusion) compared to the backing ensemble. This is done by the use of a combination of reverberation and artificial balance. Just like the larger than life presence that movie stars have on the cinema screen, popular singers in recordings can have a god-like presence. They can sing with a power not possible in the real world and, due to recording and broadcasting technology, defy time and space.

These illusions cause music to be heard by the listener as distinct from the mundane or everyday listening experience. Indeed the spatial imagery commonly found in popular recordings creates a spatial context for music that is unreal or even surreal. Listeners accept the unreality of music heard through the stereo sound system in the same way in which they are able to accept the unreality of a puppet show – by a kind of aural suspension of disbelief. Kendall (1995) describes recorded music as a

> “learned cultural form that we usually take to be a mediated approximation to direct experience”.

Unreal illusions of space can provide many creative approaches to the composition of music. They may, however, reinforce an unconscious association in which the sound that is being heard is distinct from the mundane and removed from the listener’s personal space. This facilitates the commodification of music

and sound by the listener. The recording industry can be seen as producing objects that are consumed the same way as any other mass-produced good — machine perfect and not able to be made by individuals. This can give rise to the speculation that the everyday immersive experience of sound has diminished as an aesthetic experience for a large number of people. Support for this can be seen it the fact that acoustic spaces, such as shops, cars, workplaces and homes are increasingly filled with the sound of radios and other playback equipment. Here everyday experience is masked by a spatial illusion.

Multi-speaker Systems and Acoustic Space

Since the 1950's composers have experimented with multi-speakers systems. These experiments have given rise to approaches that utilise the acoustic space in which the loudspeakers are in. An early example of this is Karlheinz Stockhausen's *Gesang der Jünglinge* (1956) which is a piece for tape consisting of 5 tracks distributed to 5 loudspeakers according to a predetermined plan (Stockhausen, 1961). Another notable piece from the same period is *Williams Mix* (1953) by John Cage, a work for eight channels of audio. Here the spatial distribution is determined by Cage's chance operations (Chadabe, 1997 pp. 56 -58).

In 1974, François Bayle created the *Acousmonium*, which consisted of an orchestra of 80 loudspeakers. Bayle placed the loudspeakers on a stage arranged in a manner similar to that of an orchestra in that the instruments where placed according to their range, power and other projecting qualities (Bayle, 1986). This method of presentation has become a popular model for composers utilising multi channel sound.

Another approach is seen in the *Moving Sound Creatures* created by Felix Hess in 1987. Hess created 24 simple robots that were mobile and produced sounds. Together the robots create a kind of dance that results in a continually changing spatial distribution of sound (Chadabe, 1997).

The works discussed above overcome the problem of directionality in loudspeakers by using multiple speakers, thereby projecting the sound in a way that is similar to the way in which ensembles of instruments do. In addition they also overcome the problem of superimposed spatial imagining between virtual and acoustic spaces as there is no intention to create a virtual spatial illusion in the way that occurs with the stereo sound system. Instead space is created in the acoustic space of the performance by the use of multiple speaker arrays and the animation of sounds occurring in those arrays.

The development of multi-speaker techniques has given rise to an approach called *Sound Diffusion*. Here composers such as Harrison[4] distribute a limited number of channels, via a mixing console, to a larger number of loudspeakers. This method of spatial distribution has been expanded further with the introduction of software that automates the process of distributing the sound. An example of this is the ABControl software interface that controls the Richmond AudioBox which was recently used at the Sonic Residues concerts in Melbourne last year[5].

While this approach affords many options for composers working in this area it can fall short of creating a convincing sonic landscape. If Moore's concept of spatial imaging in loudspeakers is taken into account this criticism becomes apparent. In Moore's model, loudspeakers function like windows that open up to a world of sound outside the space in which the speakers occupy. Therefore a single sound that is assigned to one loudspeaker in a multi-speaker array is not convincing unless it is distributed to all of the loudspeakers with corresponding alterations in onset times and intensity that allow for the difference in paths taken for sound to reach the respective loudspeaker windows.

The solution to these problems requires the creation of virtual acoustic spaces. There is now a large body of literature on this subject that outlines a range of approaches to the creation of virtual space. However

[4] During the mid 1980's Jonty Harrison and colleagues at the Birmingham University developed BEAST (Birmingham Electro Acoustic Sound Theatre) which utilises a complex assortment of loudspeakers for sound diffusion (Chadabe 1997 p. 132).

[5] For further information about the ABControl and the Richmond AudioBox software see: http://www.thirdmonk.com

this literature tends to be highly technical making it somewhat unapproachable for many composers. For this reason the discussion here will cover some of the approaches in conceptual terms oriented towards a compositional perspective.

Virtual Spaces and Loudspeakers

Virtual spaces are created by playing sounds which have spatial cues present in the signals. In the case of the stereo sound system, this is done by recording sound with two microphones that record spatial cues created by the environment as part of the signal. This method can be extended to multi-channel recording when more than two microphones are used. An example of this is the Ambisonic system for the recording and playback of sound. This system not only records right and left information (using the coincident pair method used in stereo) but also front and back, and height[6].

Another method is to subject a sound to signal processing that modifies the signal to include spatial cues. An early example of this, in which a computer was used to simulate moving sounds in a quadraphonic sound system, was developed from 1966 onwards by John Chowning. Here Chowning developed a computer program that calculates a range of spatial cues (such as relative energy levels, reverberation and doppler shifts) necessary for moving sounds within a virtual space (Chowning, 1971). After Chowning's work, a wide range of approaches, using the signal processing capabilities of computers to create virtual spaces, have been developed.

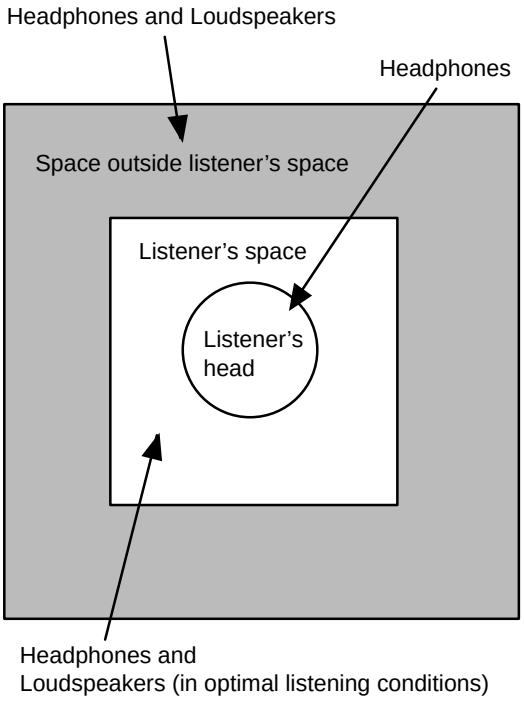


Figure 2. Three possible spaces that can be created using either headphones or loudspeakers.

[6] See Malham (1998) for an introduction to Ambisonic technology.

Virtual spaces can be categorised into three basic types (see Figure 2). They are the space:

- outside the listener's acoustic space,
- inside the listener's acoustic space, and,
- inside the listener's head.

The boundaries of these spaces are determined by the position of the loudspeakers. In the first and second categories, the space exists either outside or inside the boundary defined by the positions of the loudspeakers. In the third case, it is delimited by the position of the loudspeakers in the headphones.

The first category is addressed by Moore's system for spatialising sound. Moore's algorithm relies on the idea that loudspeakers are windows into a space outside the listener's space. It is a robust method because it not distorted by the acoustic space. This is because the spatial illusion is heard outside, and therefore distinct from, the acoustic space. Moore's approach however, is not able to create illusions for sounds positioned inside the listener's acoustic space.

Chowning's approach, and similar approaches after Chowning, attempt to position and move sounds within the listener's space[7] (the second category described here). When signals processed in this way are played over loudspeakers however the problem of the overlay of virtual and acoustic spaces becomes a factor. For this reason, loudspeaker systems, such as the speaker dome developed at ACAT, must be used in an environment in which reflected sound is kept to a minimum (Vennonen, 1995).

This problem is avoided if loudspeakers are replaced with headphones. However this gives rise to another problem. If the extent of the virtual space is now between the loudspeakers, then this space must be, to the listener, inside the head. While there is some novelty in hearing sounds in this way (although at times the sensation can be somewhat disconcerting) there is still a failure in representing sound as occurring in the acoustic space of the listener. If the sounds are positioned so that they are perceived outside the bounds of the headphones then we are returned to Moore's model of spatialisation.

In terms of an immersive sound field, headphones are problematic for several reasons (see Kendall, 1995 pp. 37-39). One of these reasons is that if the listener moves his or her head the virtual image will move also. The result of this correlation of movement is the conclusion that the sound must be inside that head.

With the advent of the DVD standard for the domestic market (Dennis, 2000) there has been a rapid growth in software that performs spatial processing[8]. A survey of this new growth in electronic music is well beyond the scope of this paper.

Space as a Musical Parameter

As discussed above, there have been a number of approaches to incorporating spatiality into the distribution process. However, in the compositional process, spatiallity has tended to be a secondary consideration, or at least a consideration that is imposed or overlayed after the other compositional parameters have been set. It therefore seems timely that the development of techniques for the composition of spatial parameters be addressed.

An important contribution towards incorporating spatiality into the compositional process can be found in the writings of Wishart on the subject of spatial motion (1996, pp 191 – 235). Here Wishart catalogues a wide range of trajectories that could be used for moving sounds in a two dimensional plane. While this kind of analysis is useful, Wishart does not mention how his catalogue of trajectories might have a

[7] These approaches can also position sounds outside this space.

[8] Examples of this can be seen in the plugins being used by audio and MIDI sequencing software such as Digital Performer (Mark of the Unicorn, 2001). These programs have incorporated mixing facilities that enable mastering to DVD audio specifications.

correlation with other musical parameters. Furthermore, he stops short of discussing compositional methods or approaches to the long-term organisation of the trajectories he presents.

An approach that extends the idea of space as a compositional parameter is found in the author's SNet software (McIlwain & Pietsch, 1996). This approach attempts to incorporate the spatial parameter in such a way that it becomes intrinsic to the compositional process. Here the spatial parameter is considered in the same manner as are the commonly considered parameters of frequency, amplitude and time. The consideration of these three parameters in the compositional process is usually interrelational, that is, depending on context, a composer will base decisions about one of these parameters in the context of the other two. This is done in SNet by the use of a nodal network in which all of the parameters influence, in an interrelational manner, the way the software generates music.

Conclusion

This paper has attempted to show aspects of the distinction between virtual and acoustic spaces in terms of loudspeaker use. Virtual space, with currently available technology, can be compared to film. Here a convincing visual illusion is created that incorporates three dimensions. The audience however, is unable to walk into the film and experience it as a three dimensional event. Like the screen in the cinema, the loudspeaker is the barrier to virtual space. There are currently developments in virtual reality that enable users to walk around in computer-animated spaces and it is conceivable that this might be possible in a sound landscape also. A complete solution to this problem may require the development of new technology that can project illusory sound sources into the listening space. For the time being it seems that the model in which loudspeakers are holes into a sound world outside that of the listener prevails.

This is not to suggest that the spatial limitations in relation to loudspeakers necessarily limit the creative possibilities for electronic music. However a classification of virtual spaces, which this paper attempts to outline, and an understanding of the perceptual relationship of the listener to these spaces, may be useful for the development of sophisticated compositional technique.

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